

Nonvanishing of X_3 in Bigelow's Zipper-Algebra Presentation

A Burau Witness for X_3 and a Finite-Dimensional Quotient Above BMW

Abstract

Bigelow's Question 6 asks whether the recursively defined element X_3 vanishes in the zipper-algebra quotient. This note gives a self-contained exact calculation showing that it does not vanish under either natural well-typed interpretation of the printed relation scheme. An unreduced Burau representation yields the uniform identity

$$\rho(X_k) = \left(\prod_{j=1}^{k-1} (q^j - u) \right) E_{k-1}, \quad \rho(X_3) = (q - u)(q^2 - u)E_2 \neq 0,$$

where E_{k-1} is an explicit nonzero rank-one matrix. All defining relations are verified symbolically, including both natural repairs of a terminal indexing defect in the printed presentation. The same construction is compatible with $\sigma_1 X_2 = tX_2$ after the specialization $u = -t$ and produces a finite-dimensional quotient that maps onto the BMW algebra with nonzero kernel. The final sections delimit the scope of these conclusions and give an exact test for compatibility with further proposed relations.

1 Introduction and main results

The zipper algebra was introduced as a braid-group-algebra quotient intended to retain more information than the BMW algebra while remaining amenable to a finite-dimensional description. Question 6 asks whether the element X_3 already vanishes in the defining quotient. Vanishing would collapse the relevant branch to the BMW quotient; nonvanishing is therefore the first structural test for a genuinely larger algebra.

The main result is an explicit generic witness for nonvanishing. The unreduced Burau representation sends every recursively defined X_k to a nonzero rank-one direction multiplied by a known scalar polynomial. In particular,

$$\rho(X_3) = (q - u)(q^2 - u)E_2 \neq 0$$

over $\mathbb{Q}(q, u)$. The calculation is uniform in n , verifies every well-typed displayed relation, and remains valid under either natural repair of the terminal indexing defect in the printed presentation.

Two additional consequences are established. First, after setting $u = -t$, the witness satisfies $\sigma_1 X_2 = tX_2$ without killing X_3 . Second, the Burau image can be combined with the standard BMW quotient to produce a finite-dimensional quotient that surjects onto BMW with nonzero kernel. This gives a precise finite-dimensional proper extension in the quotient-extension sense. It does not assert that the universal zipper algebra itself is finite-dimensional or provide a canonical diagrammatic basis.

2 The repaired presentations

Bigelow works over a domain R with invertible parameters. The nonvanishing statement is made over the generic field used below (equivalently, after generic base change). This is enough to disprove

a universal identity: if X_3 were zero in the presented algebra, every representation after every base change would send it to zero.

Bigelow defines Z_n as a quotient of RB_n , but the final displayed row uses σ_n , which is not a generator of B_n . Let Z_n^* denote either of the following two explicit presentations:

- (i) retain in RB_n every displayed relation whose braid generators actually occur, namely the rows through the last legal index;
- (ii) place the terminal row in RB_{n+1} , equivalently interpret the displayed rows as a compatible direct-limit relation scheme.

No equality between these repaired quotients is assumed. The representation below separately satisfies all relations in each presentation. Thus the terminal-index defect does not affect the answer to Question 6 under either natural interpretation. Selection of one repaired presentation as the canonical definition is logically separate from the nonvanishing argument.

Bigelow's whole-word inverse convention is

$$\bar{\sigma}_{i_1 \dots i_k} = \sigma_{i_1 \dots i_k}^{-1}.$$

In particular, $\bar{\sigma}_{21} = (\sigma_2 \sigma_1)^{-1} = \sigma_1^{-1} \sigma_2^{-1}$.

The recursively defined elements are

$$\begin{aligned} X_2 &= q\bar{\sigma}_1 + 1 - q - \sigma_1, \\ X_k &= \left(q^{k-1} \bar{\sigma}_{(k-1) \dots 1} - \sigma_{1 \dots (k-1)} \right) X_{k-1}. \end{aligned}$$

3 Exact Burau calculation

Work first over $K = \mathbb{Q}(q, u)$ and let $V = K^n$ with standard basis $e_1, \dots, e_n$. Send σ_i to the unreduced Burau matrix B_i whose only nonidentity block, on coordinates $i, i+1$, is

$$B_i|_{i, i+1} = \begin{pmatrix} 1-u & u \\ 1 & 0 \end{pmatrix}, \quad B_i^{-1}|_{i, i+1} = \begin{pmatrix} 0 & 1 \\ u^{-1} & 1-u^{-1} \end{pmatrix}.$$

These matrices satisfy the Artin braid relations and $\det(B_i) = -u \neq 0$.

For $1 \leq r < n$, define

$$a_r = u^{1-r} e_r - u^{-r} e_{r+1}, \quad b^T = -e_1^T + e_2^T, \quad E_r = a_r b^T.$$

Each E_r is a nonzero rank-one matrix.

Lemma 1 (Local transport identities). *For every admissible $r \geq 2$,*

$$\begin{aligned} (B_1 \cdots B_r) a_{r-1} &= u a_r, & (B_r \cdots B_1)^{-1} a_{r-1} &= a_r, \\ (B_2 \cdots B_{r+1}) a_r &= u a_{r+1}, & (B_{r+1} \cdots B_2)^{-1} a_r &= a_{r+1}. \end{aligned}$$

The same identities hold after replacing a_r by E_r , since all operators act by left multiplication.

Proof. They follow by substituting the two-by-two block formulas. After the only nontrivial adjacent blocks have acted, the resulting vector is supported on the final two coordinates, so all remaining lower-indexed blocks fix it. $\square$

Theorem 1 (Uniform formula and nonvanishing). *For every $2 \leq k \leq n$,*

$$\rho(X_k) = \left(\prod_{j=1}^{k-1} (q^j - u) \right) E_{k-1}.$$

The representation factors through each repaired presentation $Z_n^\star$. Consequently,

$$\boxed{\rho(X_3) = (q - u)(q^2 - u)E_2 \neq 0,}$$

and therefore $X_3 \neq 0$ in each repaired presentation for every $n \geq 3$.

Proof. Direct multiplication gives

$$\rho(X_2) = qB_1^{-1} + (1 - q)I - B_1 = (q - u)E_1.$$

The first pair of transport identities and Bigelow's recursion then prove the closed formula by induction.

For the exceptional first defining relation, a direct 3×3 calculation gives

$$(qB_2^{-1} + (1 - q)I - B_2) E_1 = (q(B_2B_1)^{-1} - B_1B_2) E_1 = (q - u)E_2.$$

For every later row, the shifted transport identities give

$$(q^{k-1}(B_k \cdots B_2)^{-1} - B_2 \cdots B_k) E_{k-1} = (q^{k-1} - u)E_k,$$

and the expression with the additional B_1 and B_1^{-1} has the same value. Thus both sides of every legal displayed relation have the same image. The identical local calculation also handles the terminal row when it is placed in B_{n+1} or the direct limit.

Finally, $q - u$ and $q^2 - u$ are nonzero in K , and E_2 is nonzero. Hence $\rho(X_3) \neq 0$. $\square$

Conceptually, B_1 has eigenvalues 1 and $-u$. The element X_2 kills the 1-eigenspace and is a scalar multiple of the rank-one projector onto the $-u$ direction. The recursion defining X_k simply transports that direction along adjacent strands. This is why the apparently long family of zipper relations collapses to the four local transport identities above.

The witness is generic. At a special fiber such as $u = q$ or $u = q^2$ its displayed image of X_3 degenerates, and this particular representation then makes no nonvanishing claim for that specialization.

For reference, the nonzero block of the witness is completely explicit:

$$\rho(X_3) = (q - u)(q^2 - u) \begin{pmatrix} 0 & 0 & 0 & \cdots \\ -u^{-1} & u^{-1} & 0 & \cdots \\ u^{-2} & -u^{-2} & 0 & \cdots \\ 0 & 0 & 0 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

4 The relation $\sigma_1 X_2 = tX_2$

This section tests the Burau witness against the natural additional relation $\sigma_1 X_2 = tX_2$, considered in finite-dimensional refinements of the zipper algebra. The vector $a_1 = e_1 - u^{-1}e_2$ satisfies

$$B_1 a_1 = -u a_1.$$

Since $\rho(X_2) = (q - u)E_1$, it follows that

$$\boxed{\rho(\sigma_1 X_2) = -u \rho(X_2).}$$

After setting $u = -t$ and working over $K_t = \mathbb{Q}(q, t)$, the Burau representation therefore satisfies

$$\rho_t(\sigma_1 X_2) = t \rho_t(X_2),$$

while

$$\boxed{\rho_t(X_3) = (q + t)(q^2 + t)E_2 \neq 0.}$$

Thus imposing the eigenvalue relation $\sigma_1 X_2 = tX_2$ does *not* force $X_3 = 0$. If a larger collection of proposed relations forces $X_3 = 0$, then at least one of the following must occur:

- (a) one of those relations has nonzero image under ρ_t ;
- (b) the calculation uses a parameter specialization at which the generic Burau witness degenerates;
- (c) a different inverse, indexing, or coefficient-ring convention is in use;
- (d) the additional relations genuinely remove the Burau branch.

For any specified relation set and parameter regime, these alternatives can be distinguished by exact symbolic evaluation.

5 Historical and current-work context

Three records need to be read together.

- (1) A 2004 BIRS workshop report described Bigelow's zipper algebras as a finite-dimensional proper extension of BMW. This is a secondary workshop report rather than a published proof of nonvanishing.
- (2) The 2005 preprint, published in 2006, explicitly defines the original quotient Z_n , asks whether $X_3 = 0$, and separately asks which extra relations make Z_n finite-dimensional. Thus the published source itself leaves both questions open.
- (3) Bigelow's AMS abstract for 4 May 2025, *The zipper algebra*, proposes a finite-dimensional algebra having Temperley–Lieb, Hecke and BMW as quotients, with a finite normal-form algorithm, and identifies it as current work with Choomno Moos. The abstract does not give the defining relations or state that Question 6 for the original 2005 presentation has been settled.

The conclusion established here is therefore specific and checkable: the original displayed Z_n relation scheme, under either natural repair of its terminal index, admits the Burau representation and hence does not kill X_3 . Whether a later finite-dimensional zipper-algebra presentation retains this Burau branch is a separate compatibility question.

6 A finite-dimensional algebra strictly above BMW

The preceding parameter match yields a precise construction addressing the original motivation.

Let

$$Y_n = Z_n^* / \langle \sigma_1 X_2 - t X_2 \rangle$$

over $K_t = \mathbb{Q}(q, t)$. Bigelow's standard quotient, obtained after also imposing $X_3 = 0$, gives a surjection

$$\pi : Y_n \twoheadrightarrow \text{BMW}_n(K_t).$$

The specialized Burau representation gives

$$\rho_t : Y_n \longrightarrow A_n \subseteq M_n(K_t),$$

with $\rho_t(X_3) \neq 0$.

Proposition 1 (Finite-dimensional proper extension of BMW). *Define*

$$D_n = \text{im}(\rho_t, \pi) \subseteq A_n \times \text{BMW}_n(K_t).$$

Then D_n is a finite-dimensional quotient of Y_n , and projection onto the second factor induces a surjection

$$D_n \twoheadrightarrow \text{BMW}_n(K_t)$$

with nonzero kernel. In particular,

$$\dim_{K_t} D_n > \dim_{K_t} \text{BMW}_n(K_t).$$

Proof. The algebra A_n is contained in $M_n(K_t)$, so it has dimension at most n^2 over K_t . The generic BMW algebra is finite-dimensional; therefore D_n , being a subalgebra of their direct product, is finite-dimensional. The second projection is surjective because π is surjective. Moreover,

$$X_3 \longmapsto (\rho_t(X_3), \pi(X_3)) = (\rho_t(X_3), 0) \neq 0.$$

This element lies in the kernel of the second projection, proving that the kernel is nonzero and the dimension is strictly larger. $\square$

Remark 1. This proves the existence of a finite-dimensional quotient strictly above BMW in the quotient-extension sense. It does not yet give a preferred diagrammatic basis or prove that the universal algebra Y_n itself is finite-dimensional. In particular, it is not automatically the same algebra as the normal-form construction announced by Bigelow and Moos in 2025. If the intended goal is a canonical presentation with a tractable basis and terminating reduction algorithm, that remains an additional structural problem.

7 Compatibility test for additional relations

For each proposed relation $r_\alpha \in K_t B_n$, compute the exact matrix $\rho_t(r_\alpha)$.

- If $\rho_t(r_\alpha) \neq 0$, then that relation removes the present Burau branch. Its nonzero entries identify the precise obstruction.
- If every $\rho_t(r_\alpha) = 0$, then the quotient by all those relations still has the Burau image and therefore cannot force $X_3 = 0$ generically.

- If vanishing occurs only after specialization, record the exact polynomial conditions on q and t . This separates a generic collapse from a special-fiber collapse.

An exact comparison requires the following presentation data:

1. the exact braid words for every additional relation;
2. the coefficient ring and all parameter specializations;
3. the convention for $\bar{\sigma}_{i_1 \dots i_k}$;
4. the values of n used in the computer calculations;
5. whether “finite-dimensional and bigger than BMW” means a proper quotient-extension as above, or a universal algebra with an explicit basis;
6. the defining relations or normal-form rules of any revised zipper algebra, for comparison with the original Z_n .

Once fixed, these data turn the comparison into exact symbolic matrix identities rather than heuristic numerical tests.

8 Independent replay and audit trail

The accompanying program `verify_bureau.py` independently reconstructs the matrices and checks:

- the closed formula for all X_k available at each order;
- every displayed defining relation whose generators exist;
- the terminal-row calculation for the second repaired reading;
- $\text{rank}(\rho(X_3)) = 1$ and nonvanishing.

Using Python 3 and SymPy 1.14.0, the exact replay returned

```
n=3: passed; rank(X3)=1
n=4: passed; rank(X3)=1
n=5: passed; rank(X3)=1
n=6: passed; rank(X3)=1
n=7: passed; rank(X3)=1
n=8: passed; rank(X3)=1
bounded_independent_replay=passed
```

The finite replay is a regression and transcription check. The proof for all n is the local block calculation and induction above.

Current SHA-256 identifiers are:

```
fac9104a79ac54013a13258de415a0f3c67b04d674fa0d6b0880e9566c63b3ef
  BIGELOW_X3_RESULT.md
8fe24dbe9138310cd45c0dbf1d09a70e9d3587cc683213cf4660cf3d0cb40b22
  bureau-witness-proof.md
4343e1272911b12fe16c16b1d63964c9d79ae3bde63fa3a10c2ddb130621cc19
  verify_bureau.py
```

9 Scope and unresolved comparisons

The published record does not determine three pieces of information required for a complete comparison with later variants of the presentation.

1. **Presentation convention.** The correction to the terminal indexed relation that matches the intended definition of Z_n .
2. **Additional relations.** The exact relations used in subsequent finite-dimensional experiments, together with the parameter assumptions and the smallest order at which the collapse $X_3 = 0$ was observed.
3. **Current presentation.** The relations or reduction rules for the zipper algebra announced in the 2025 AMS abstract, if that construction uses a revised presentation.

With these data fixed, the comparison reduces to an exact calculation: evaluate the complete relation set in the generic Burau image, identify the first relation that removes it, and determine whether the finite-dimensional algebra D_n above realizes the intended notion of “bigger than BMW.”

Next mathematical steps

1. **Freeze the intended presentation.** Record the confirmed correction of the terminal index and the exact additional relations as a machine-readable list of elements of the braid-group algebra.
2. **Run the obstruction test.** Compute every $\rho_t(r_\alpha)$ symbolically, factor all nonzero entries, and produce a minimal subset of relations that kills the Burau branch. Repeat over generic parameters and over every specialization used in Choom’s experiments.
3. **Search beyond Burau if necessary.** If an intended relation has nonzero Burau image, solve the same relation constraints in larger representation families (reduced Burau, Lawrence–Krammer, Hecke/BMW module extensions, and low-dimensional exact matrix ansatzes) while retaining a nonzero image of X_3 .
4. **Make the finite-dimensional quotient structural.** Compute the dimension, kernel, generators, and a basis of D_n for small n ; then look for a uniform diagrammatic or rewriting presentation rather than only the abstract product-image construction.
5. **Independent verification.** Have a braid-algebra specialist check the source conventions and the hand proof, and have a second system replay the exact matrices from a clean environment. Archive the scripts, outputs, and hashes with the note.
6. **Write the publishable note.** State the corrected presentation, the uniform nonvanishing theorem, the comparison with Choom’s relations, and the finite-dimensional quotient result. The introduction should explicitly credit Bigelow’s unpublished 2004 proof attempt and distinguish independent rediscovery from historical priority.

References

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- [4] J. Birman, P. Dehornoy, and D. Rolfsen, *BIRS workshop on braid groups and applications*, report of BIRS workshop 04w5526, 2004, pp. 3–4.
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